**Congratulations for seeking Govt. Sponsored Project**

Name: Dr. Aftab Alam (Principal Investigator)

Name: Dr. S. Shiva Kumar (Principal Investigator)

Department: Electrical and Electronics Engineering

Project Title: **Development of High-Power Grid-Friendly Conductive and Static Wireless Chargers for Electric Vehicles. (Work package – 2 High Power Density Static Wireless Charger**

Funding Agency: ANRF (under EV Mission)

Sanction Amount: 150.00 Lakhs

Tenure: 3 Years

Used amt.: Rs. 8888831

Abstract:

Wireless Power Transfer (WPT) has emerged as a transformative technology enabling the delivery of electrical energy without physical connectors, offering significant advantages in safety, convenience, and system reliability. In this field, under the MAHA-EV Mission, ANRF has sanctioned a grant of 1.45 Cr to Dr. Aftab Alam and Dr. Shiva Kumar of the Electrical and Electronics Engineering Department at Birla Institute of Technology, Mesra. The major objective of this project is to develop a 50kW Wireless Power Transfer (WPT) system for charging Electric Vehicles (EV). The WPT system sends power from a transmitting coil embedded in the floor to a receiving coil at the bottom surface of the EV in order to charge the onboard battery of the EV. However, this is a very complex process due to the inefficient electromagnetic coupling between the coils through the airgap. This challenge will be addressed in this project by developing an innovative coil design and a coil excitation system. Also, the other major challenge is the reduced performance of WPT due to misalignment between the transmitting coil and receiving coil due EV driver error in parking vehicle exactly over the transmitting coil. This issue will also be addressed in this project. Currently, there are no indigenous solutions available for commercial deployment at this power level.